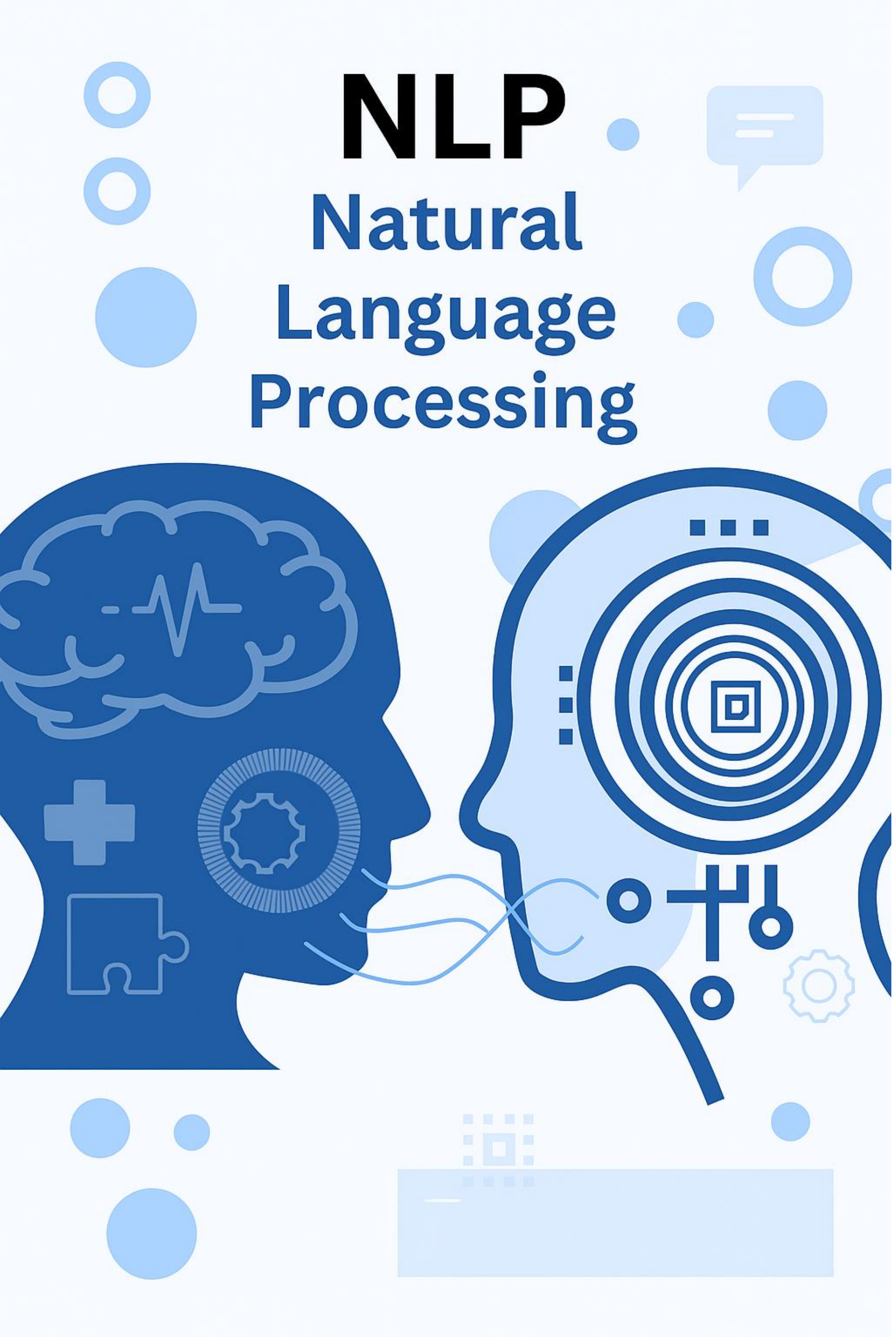

NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 4

LECTURE 5

Dr. Kamanasish Bhattacharjee

Assistant Professor

Dept. of Analytics, SCOPE, VIT

LEVEL 2: MORPHOLOGICAL LEVEL

- Deals with: Structure of words (morphemes — smallest written units)
- Goal: Analyze and generate word forms.
- Example: “unbelievable” → un + believe + able
- Applications: Lemmatization, stemming, spell checking

Morphology

Morphology studies the internal structure of words, how words are built up from smaller meaningful units called **morphemes**.

Morpheme: It is minimal meaning bearing unit in a language.

Examples:

- dogs: 2 morphemes- ‘dog’ and ‘s’.
  ‘s’ is a plural marker on nouns.
- Unladylike: 3 morphemes
  un- ‘not’
  lady- ‘well-behaved woman’
  like- ‘having the characteristic of’ ns.
- fox: 1 morpheme (singular); foxes: 2 morphemes (plural)

Morphology

Free morphemes – can stand alone as words.
Example: book, run, happy

Bound morphemes – cannot stand alone; attach to other morphemes.
Example: prefixes (un-, re-) and suffixes (-ed, -ing, -s).

Morphology

Classes of Morphemes

Two broad classes of morphemes are:

- Stems – the core meaningful units, the root of the word.
- Affixes – bits and pieces adhering to stems to change their meanings and grammatical functions.
    - ***Prefixes*** – precede the stem: do / undo
    - ***Suffixes*** – follow the stem: eat / eats
    - ***Infixes*** – are inserted inside the stem
    - ***Circumfixes*** – precede and follow the stem

Morphology

Affixes – Prefix vs Suffix

Prefix

Word or part of word added before stem or root. Examples:

- bi (two): bicycle, bilingual
- dis (not): dishonest, disagree, disapprove, disconnect
- in (not, without, lacking): injustice, incomplete, invisible
- ir (not, non, opposite) :irreversible, irreplaceable, irregular
- More common prefix example

Suffix

Word or part of word added after stem or root. Examples:

- able (can be done): preventable, adaptable, predictable, credible
- er (some one who performs an action): teacher, helper, preacher, dancer
- -ion (the action or process of): celebration, opinion, decision, revision
- More common suffix example

Morphology

Affixes – Infix vs Circumfixes

INFIX

Word or part of word added in between the stem or root word.

Examples:

- passersby: plural of passerby
- mothers-in-law : plural of mother-in-law
- More common in informal English.

CIRCUMFIXES

Circumfixes attach to both the beginning and the end of a word

Examples:

- in-adequate-ness
- in-correct-ness
- im-perfect-ion
- less common in English.

Morphology

Affix Types

Affix type	How it attaches to stem
Prefix	X-STEM
Suffix	STEM-X
Infix	ST-X-EM
Circumfix	X-STEM-X

Morphology

There are two broad (and partially overlapping classes of morphology

1) Inflectional Morphology
2) Derivational Morphology

INFLECTIONAL MORPHOLOGY

Grammatical: Add the grammatical property to the word like tense, number (plural), case, gender, comparison

Creates new forms of the same word: bring, brought, brings, bringing.

DERIVATIONAL MORPHOLOGY

Creates new words by changing part-of-speech: logic, logical, illogical, illogicality, logician

Fairly systematic but some derivations missing: sincere - sincerity, scarce - scarcity, curious - curiosity, fierce - fiercity?

Morphology

Inflectional Morphology

- Inflectional morphology deals with **grammatical variations of the same word**. It changes the form of a word **without changing its core meaning or category** (noun stays noun, verb stays verb).
- It shows **tense, number, gender, case, aspect, or degree**.

Examples:

- Noun Inflection:
  - **cat** → **cats** (plural)
- Verb Inflection:
  - **walk** → **walked** (past tense)
  - **walk** → **walking** (progressive aspect)
- Adjective Inflection:
  - **fast** → **faster** → **fastest** (comparative and superlative)

- Inflectional analysis is crucial for **lemmatization** (finding the dictionary/base form of words) and **POS tagging**, where the system must recognize that *walked* and *walking* are forms of *walk.*

Morphology

Derivational Morphology

- Derivational morphology creates **new words** by adding morphemes (prefixes/suffixes) that often **change the meaning and sometimes the grammatical category** of the base word.
- Unlike inflection, derivation can form **completely new lexical items**.

Examples:

- Noun → Adjective:
  - **nation → national**
- Verb → Noun:
  - **teach → teacher**
- Adjective → Adverb:
  - **happy → happily**
- Prefix changes meaning:
  - **possible → impossible**

Derivational analysis helps in **word formation rules, vocabulary expansion, and semantic interpretation**. For example, knowing that *happiness* is derived from *happy* helps systems cluster related concepts in text mining or sentiment analysis.

Morphology

Inflectional Morphology vs Derivational Morphology

INFLECTION

1. An *inflectional morpheme* is added to a noun, verb, adjective or adverb to assign a particular **grammatical property** to that word such as: tense, number, possession, or comparison.
   - Cat $\Longrightarrow$ Cats, Teach $\Longrightarrow$ Teaches
   - Clean $\Longrightarrow$ Cleaned, Pretty $\Longrightarrow$ Prettier

DERIVATIONAL

1. *Derivational morphemes* form a new words that differ either in grammatical category (parts-of-speech) or in meaning from their bases.
   - **Change in Meaning**: Leaf $\Longrightarrow$ leaflet, Pure $\Longrightarrow$ Impure
   - **Change in Grammatical Category**: Help (verb) $\Longrightarrow$ Helper (noun), Logic (noun) $\Longrightarrow$ Logical (adjective)

Morphology

Inflectional Morphology vs Derivational Morphology

INFLECTION

2. These do do not change the essential meaning or the grammatical category (Parts-of-speech) of a word. Adjectives stay adjectives, nouns remain nouns, and verbs stay verbs.
   - Clean (Verb) $\Longrightarrow$ cleaned (Verb)
   - Cat (Noun) $\Longrightarrow$ Cats (Noun)

DERIVATIONAL

2. *Derivational morphemes* tend to change the grammatical category of a word **but not always**!
   - ful like in 'beautiful' $\Longrightarrow$ beauty (N) + ful (A) = beautiful (A)
   - able like in 'moldable' $\Longrightarrow$ mold (V) + able (A) = moldable (A)
   - -er like in 'singer' $\Longrightarrow$ sing (V) + er (N) = singer (N)

Morphology

Inflectional Morphology vs Derivational Morphology

INFLECTION

3. In English, all inflectional morphemes are suffixes (i.e. they all only attach to the end of words).
4. There can only be one inflectional morpheme per word
5. They create new form of same word.

DERIVATIONAL

3. There can be multiple derivational morphemes per word
4. They can be prefixes, affixes, or suffixes
5. They create new words.

Morphology

Inflectional Morphology vs Derivational Morphology

INFLECTION

Examples:

- Plural: Bikes, Cars, Trucks, Lions, Monkeys, Buses, Matches, Classes
- Possessive: Boy's, Girl's, Man's, Mark's, Robert's, Samantha's, Teacher's, Officer's
- Tense: cooked, played, marked, waited, watched, roasted, grilled; sang, drank, drove
- Comparison: Faster, Slower, Quicker, Taller, Higher, Shorter, Smaller, Weaker, Stronger, Sharper, Bigger
- Superlative: Fastest, Slowest, Quickest, Tallest, Highest, Shortest,

DERIVATIONAL

Examples:

- Ful: Beautiful, Wonderful, Cheerful, Truthful, Tasteful, Flavourful, Joyful
- Able: Walkable, Understandable, Loveable, Laughable, Eatable
- Ment: Government, Establishment, Agreement
- Ness: Kindness, Truthfulness, Carelessness, Sadness
- Ly: Happily, Kindly, Fortunately, Sadly, Lively